

MUSTAFA IQBAL

AI Automation & Full-Stack Engineer

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Professional Summary

AI Automation and Full-Stack Engineer building LLM-powered products, RAG systems, APIs, and operational workflows. Hands-on with Python/FastAPI, TypeScript/Next.js, vector search, OAuth, webhooks, background jobs, and database-backed SaaS systems. Focused on reliable delivery: structured outputs, sensible fallbacks, and maintainable architecture.

Core Expertise

AI Systems: LLM integrations, RAG, embeddings, vector search, multimodal AI, LangChain, prompt/workflow design, structured outputs
Engineering: FastAPI, Node.js, REST APIs, OAuth2, webhooks, Docker, PostgreSQL, Supabase, React, React Native/Expo

Experience

Independent AI Automation & Full-Stack Engineer
Freelance

2022 - Present
Remote

- Built SaaS-style automation systems for email categorization and review-response workflows, combining LLM reasoning with deterministic rules and database-backed state.
- Developed Next.js, TypeScript, Prisma, and Inngest workflows for event-driven processing, scheduled jobs, retryable background work, and webhook synchronization.
- Integrated Gmail and Google Business Profile APIs through OAuth2, supporting automated triggers, secure authentication, and ongoing data synchronization.
- Implemented hybrid AI classification with OpenAI embeddings, similarity search, confidence-aware routing, and LLM fallback for ambiguous cases.

AI Research Intern, Team Lead
Genesys Research Lab

Jun 2025 - Aug 2025
Islamabad, Pakistan

- Led a four-person team building an automated recruitment platform for AI-assisted candidate evaluation, matching, and reduced manual screening work.
- Engineered RAG retrieval over candidate and role data using Weaviate and Nomic embeddings; designed a structured conversational interview workflow using Llama 3 via Groq.
- Built modular FastAPI services with asynchronous processing and deployed containerized components using Docker Compose.

Selected Projects

MelodyMind | Multimodal AI Music Companion

Final-Year Project, 2025 - 2026

- Co-built a mobile-first AI product that turns text, image, and voice input into context-aware playlist experiences, including a live “Talk to Your DJ” interaction flow.
- Built a FastAPI architecture with Supabase/PostgreSQL, SQLAlchemy, Alembic, Pinecone, LangChain agents, health checks, JWT/refresh-token handling, and image/voice services.
- Developed a React Native/Expo client and a CLAP-InfoNCE approach aligning frozen CLAP audio features with Nomic text embeddings for emotion-aware retrieval.

AI-Powered Email Categorization Platform Gmail workflow using OpenAI embeddings, confidence-aware LLM fallback, webhooks, background jobs, and feedback loops.

AI-Powered Review Response Platform Google Business Profile automation with OAuth2, real-time synchronization, prompt templates, multi-location handling, and scheduled bulk processing.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Bash, C++, C#

AI/ML: OpenAI APIs, Gemini, Groq/Llama 3, RAG, embeddings, semantic search, LangChain, Weaviate, Pinecone, FAISS, CLAP, InfoNCE

Backend/Data: FastAPI, Flask, Node.js, Next.js, REST APIs, OAuth2, JWT, webhooks, Inngest, PostgreSQL, Supabase, MongoDB, Prisma, SQLAlchemy, Alembic

Frontend/Cloud: React, Next.js, React Native/Expo, Tailwind CSS, Docker, Docker Compose, GitHub Actions, Vercel, Render, Git, Claude Code, Codex

Education

B.S. Computer Science, FAST-NUCES, Islamabad

2026 | CGPA: **3.38/4.00** | Dean's List: **3 terms**